



ConEm: A novel framework for integrating external factors with inner and outer correlations in time series forecasting

Hoang Nguyen Nguyen^{a,*}, Wei Xiang^a, Lianhua Chi^a, Mike Da Gama^b,
Sanjeevani Avashi^c, Michael Treloar^c, Lu Yu^a

^a School of Computing, Engineering & Mathematical Sciences, La Trobe University, Bundoora, 3083, Victoria, Australia

^b Human-Kind Capital, Chirside Park, 3116, Victoria, Australia

^c NostraData, Kew, 3101, Victoria, Australia

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ABSTRACT

Time series forecasting is pivotal in both academic research and practical applications across diverse industries. However, effectively leveraging external factors to enhance forecasting performance remains a significant challenge, necessitating further investigation. Current frameworks exhibit notable limitations in modeling the impact of external factors on both intrinsic and extrinsic correlations within time series data. To address these challenges, we propose a novel mechanism that systematically integrates contextual information from external factors with temporal dependencies, while maintaining compatibility with various encoder-decoder algorithms. This approach enables backbone models to embed dependent patterns from external factors across multiple correlated time series, effectively capturing their influence on both prior and adjacent timesteps. Our study centered on the application of time series forecasting for demand prediction, as sales forecasting poses unique challenges stemming from the complexity and variability of market conditions influenced by numerous external factors. We conducted extensive experiments on three real-world retail datasets, showcasing the substantial performance enhancement of backbone models when integrated with our proposed contextual embedding mechanism. Specifically, our approach achieves improvements of up to 26% in Mean Squared Error (MSE) and 15% in Mean Absolute Error (MAE) compared to both the original backbone models and other state-of-the-art (SOTA) baseline methods. The proposed mechanism is also evaluated on Weather and Energy datasets to further verify its generalization capability. We will release the source codes and experimental datasets at our GitHub¹.

1. Introduction

Time series forecasting has emerged as a focal point of active research in both industrial and academic spheres, owing to its extensive applications across diverse domains such as retail [1], energy consumption [2], finance [3], cryptocurrency trading [4] or health care [5]. Regarding the application of time series forecasting, predicting product sales is a critical task for enterprises and has been extensively studied in numerous research projects [6–14]. Improved accuracy in sales forecasting could address numerous existing challenges within supply chain systems [9]: optimal inventory placement [15], network expansion [16], inbound and outbound planning [17], etc. A primary objective in sales forecasting is to accurately model the complex, non-linear dependencies between time series data and a diverse set of variables [10,11,18,19].

For instance, a promotion on a product can influence sales of both the promoted product and its competitor products.

The advent of sophisticated deep learning frameworks has facilitated notable advancements in the precision of time series forecasting. Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) have exhibited noteworthy outcomes by effectively capturing non-linear correlations and cross-series information [20]. Transformer architecture [21] is another viable alternative choice as it has demonstrated auspicious potential across diverse domains, encompassing fields including natural language processing (NLP) [22] and computer vision (CV) [23]. Transformers and its attention mechanism exhibit a particular aptitude for modeling extensive dependencies and interactions within sequential data, leading to the successful application of several Transformer models in the domain of time series forecasting [24–30].

* Corresponding author.

E-mail addresses: hoangnguyen.nguyen@latrobe.edu.au (H. Nguyen Nguyen), w.xiang@latrobe.edu.au (W. Xiang), l.chi@latrobe.edu.au (L. Chi), mike.dagama@outlook.com (M. Da Gama), sanjeevani.avashi@nostradata.com.au (S. Avashi), l.yu@latrobe.edu.au (L. Yu).

¹ <https://github.com/anonymous7594/conem>

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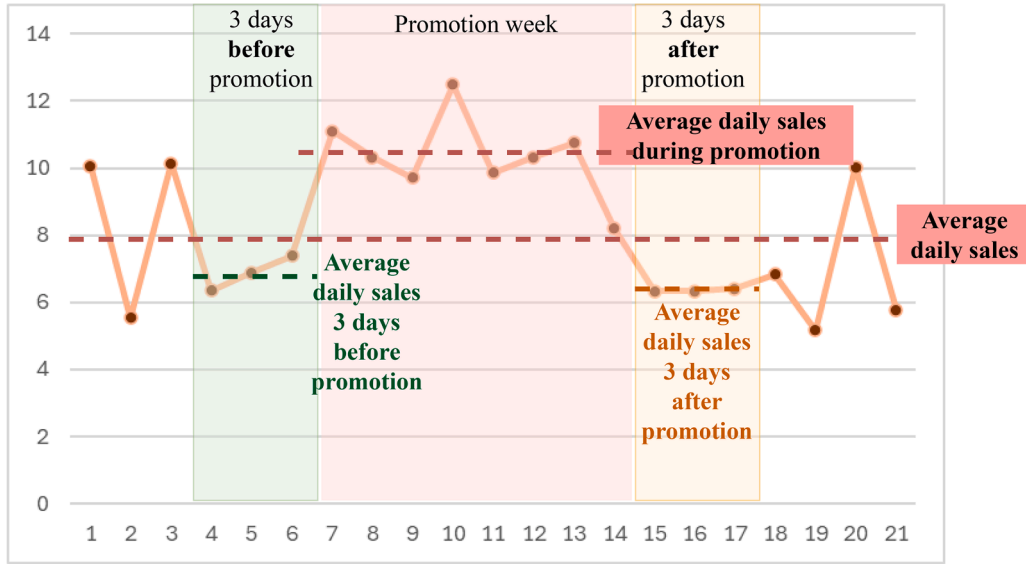


Fig. 1. Example of the average daily sales quantities of a vitamin product over the 7-day period before, during, and after its promotion week across six pharmaceutical stores in Melbourne.

Despite demonstrating favorable outcomes in numerous studies, existing deep learning frameworks face inherent limitations in incorporating external variables for time series or sales forecasting, because they predominantly focus on capturing trend and seasonality components from temporal dependencies, with insufficient emphasis on modeling the influence of external or exogenous factors [31]. We define external factors as all known variables outside the target sequences, which can be collected and integrated as additional dimensions during training and would be available in future applications to enhance forecasting accuracy. There remains a pressing need to develop more effective methods for incorporating external factors, rather than merely applying them uniformly across time periods using a global weighting mechanism in the most existing works.

It is essential to carefully analyze how the effects of external factors evolve over time and how their interactions vary across different time points. For instance, Fig. 1 shows a slight decrease in the average daily sales of a vitamin product during the time periods before and after the promotion week, as customers may delay purchases to take advantage of the promotion period, potentially leading to a decline in sales beforehand. Additionally, the increase in sales during the promotion period may affect sales in adjacent periods due to supply chain constraints or shifts in customer demand. However, most existing approaches focus solely on modeling the impact of external factors at specific time points, overlooking their effects on nearby periods.

Moreover, in terms of inner correlations, previous works have not also considered the internal factors on how it can be related to the impact of external factors. As illustrated in Fig. 2, perishable items tend to have lower sales variance compared to non-perishable products. However, the width of the distribution difference decreases as the average sales volume decreases. Therefore, information from additional factors should be incorporated when modeling the impact of external influences on each time series, as demonstrated by average sales and perishability in the example in Fig. 2. Existing deep learning frameworks primarily focus on modeling the relationships between external factors and time series, often overlooking the semantic information from internal factors. This omission limits the ability to gain deeper insights into how the impact of external factors may vary across different time series.

Regarding outer correlations, the influence of external factors on an individual time series at a specific time point can also impact its correlated time series. Modeling the external correlations between time series with and without the impact of external factors presents challenges, be-

cause external factors like promotions are irregular events. Moreover, in certain specialized industries, such as pharmaceutical retail, products may exhibit complementary or substitutive relationships due to their specific molecular compositions. Consequently, it becomes crucial to incorporate both internal and external factors associated with time series to effectively model their impact on future time series behavior. This area remains underexplored, as highlighted previously.

Our primary research objective is to propose a more efficient embedding mechanism to address the aforementioned limitations of external factor incorporation in any encoder-decoder backbone models for time series forecasting. It does so by utilising three parallel feature modules to create global embeddings for both static and time-varying factors. Subsequently, these embeddings are encoded which can be then integrated with the encoder-decoder backbone with adjustments tailored to the specific local context of the target sequence. In short, our contributions are summarized as follows:

- We propose a novel plug-and-play **Contextual Embedding mechanism (ConEm)**, designed for seamless integration into various encoder-decoder architectures. ConEm effectively captures dynamic information from both internal and external factors for each time series, facilitating the construction of global embedding layers. These global embeddings empower the model to capture the dynamic influence of external factors on both inner and outer correlations, thereby improving the accuracy of local future timestep predictions.
- We propose a specialized Feature Tokenizer block designed to decompose information from both static and time-varying factors. The Feature Tokenizer enhances the representation of relevant semantic information from both internal and external factors while maintaining a unified data representation. Our mechanism leverages this Feature Tokenizer to generate three parallel contextual embeddings, each corresponding to a distinct context.
- We propose a simple yet effective method for integrating three contextual embeddings into backbone models, enabling them to more effectively capture inner and outer correlations using known future factors. This approach also enables the identification of the influence of past factors on future patterns and utilizes static features in conjunction with recent look-back window values to adjust impact scales across both individual and correlated time series.
- An extensive evaluation of our proposed framework was conducted across both short-term and long-term forecasting settings using three

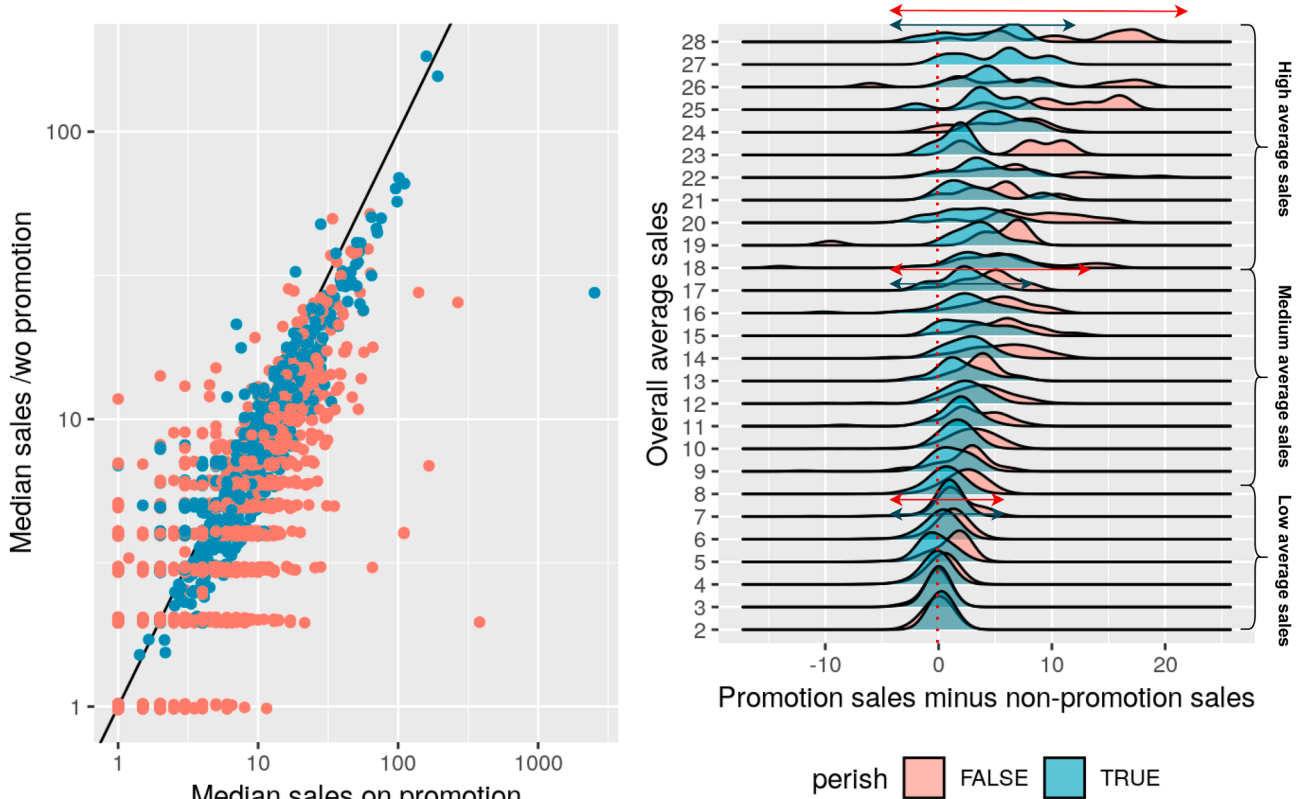


Fig. 2. Example of the sales distribution from the Corporación Favorita Grocery Sales. The chart on the right illustrates the distribution of median sales with promotions compared to median sales without promotions. The chart on the left depicts the distribution of sales differences with and without promotions across various average sales groups.

real-world retail datasets: Australian Pharmaceutical Weekly Dispensary, Australian Daily POS, and Corporación Favorita Grocery Sales. The results demonstrate a significant improvement of up to 26% in Mean Squared Error (MSE) and 15% in Mean Absolute Error (MAE) compared to backbone models and several SOTA baselines. Additional experiments on datasets from other domains, including weather and electricity price forecasting, further highlight the framework's generalization capability and its consistent accuracy improvements across different backbone architectures.

The remainder of this paper is structured as follows. [Section 2](#) reviews relevant literature, [Section 3](#) introduces notation and proposes the framework, [Section 4](#) provides the results from our experiments, and analyzes the ablation studies. [Section 5](#) provides a discussion on the limitations, applications, and directions for future research. Finally, [Section 6](#) concludes the paper.

2. Related work

2.1. Deep learning in time series forecasting

Deep learning methods can be classified into iterated approach [24] or direct approach [32]. One popular architecture network is LSTM [33], for example DeepAR [20] which applies multiple stacked LSTM layers to create one-head-ahead Gaussian predictive distributions. D³VAE [34] proposes a coupled diffusion probabilistic model to integrate the multiscale denoising score matching into the coupled diffusion process and disentangle the latent variables of the generative model. FiLM [35] redesigns the Legendre Projection Unit (LPU), makes it a general tool for data representation and proposes Frequency Enhanced Layers (FEL) that reduce dimensionality by combining Fourier analysis and low-rank matrix approximation. SDPANet [10] designs a spatiotemporal weight ma-

trix for each element of time point in multivariate time series, uses a dynamic programming algorithm to detect local stable linear correlations in a time window scanned along the timeline and provides more future information based on future trend correlations. SCINet [36] builds hierarchical down-sample-convolve-interact Time Series Forecasting framework and iteratively extracts and exchanges information at multiple temporal resolutions. TimesNet framework [37] features the TimesBlock module as a versatile backbone for conducting time series analysis, possessing the capability to dynamically discern multi-periodic patterns and effectively capture intricate temporal fluctuations within transformed 2D tensors, leveraging an inception block design optimized for parameter efficiency. The proposed seq2seq model [9] tackles the challenge of precise demand forecasting in supply chain management by introducing three deep learning models for customer sales prediction at the day/store/item level, achieving competitive performance through a sequence-to-sequence architecture with minimal preprocessing, complemented by a training technique to improve time independence and model generalization. Many recent efforts have focused on enhancing the forecasting capabilities of deep learning networks through various advanced methodologies [38–40].

2.2. MLP-based forecasting models

MLP-based frameworks have been widely proposed in the application of time series forecasting. Neural Basis Expansion Analysis for Time Series (N-BEATS) [41] contains successive block models which models the residual error due to the reconstruction of the backcast from the previous block and then updates the forecast based on that error. N-HiTS [42] proposes the enhanced version of N-BEATS with Neural hierarchical interpolation for time series forecasting (N-HiTS) to improve the treatment of input and the construction of the output. LightTS [43]

employs lightweight, sampling-based MLP architectures to decrease computational complexity and time while preserving accuracy. LTSF-Linear [44] introduces a remarkably simple one-layer linear model designed to capture temporal relationships between input and output sequences. FreTS [45] is a straightforward yet powerful architecture for time series forecasting that uses Frequency-domain MLPs, involving (i) Domain Conversion to transform time-domain signals into frequency-domain representations and (ii) Frequency Learning to capture dependencies across series by learning real and imaginary frequency components. These studies highlight the effectiveness of MLP-based networks in time series forecasting tasks and inspire the incorporation of MLPs in aspects of our work.

2.3. Transformer in time series forecasting

Transformer architectures have also been explored and got excellent results in time series [24–27,46]. LogTrans [24] proposes convolutional self-attention by applying causal convolutions on generating queries and keys in the self-attention layer, and reduces the computational complexity by a Logsparse mask. Informer [25] designs a special multi-attention mechanism and a hybrid architecture combining convolutional and transformer layers for direct method. Pyraformer [47] utilizes a hierarchical pyramidal attention module with a binary tree structure along its path to capture temporal dependencies across various ranges. Autoformer [26] uses a seasonal-trend decomposition architecture with an auto-correlation mechanism, which measures time-delay similarity between input signals and aggregates the top-k similar sub-series. FEDformer [27] enhances Autoformer architecture with frequency-domain-inspired attention operation with Fourier and Wavelet transforms, achieving linear complexity by randomly selecting a fixed size subset of frequency. Crossformer [28] proposes a Transformer model utilizing cross-dimension dependency for multivariate time series forecasting. FL-Net [29] improves long-term time series forecasting (LTSF) by using frequency external attention and multi-scale cross-decomposition to efficiently capture both trend and seasonal components. Periodformer [30] addresses the overfitting issues in LTSF caused by complex attention mechanisms, using a lightweight Period-Attention approach with a gating mechanism for enhanced periodic and proximity-based aggregation. PatchTST [48] reformulates time series forecasting as a patch-level regression task using linear projections, improving both speed and accuracy. iTransformer [49] and RLinear [50] further optimize transformer architectures by enhancing temporal saliency and reducing computational complexity, respectively. These advances highlight a growing trend toward scalable, attention-based models for efficient sequence modeling.

2.4. External factors

Although these deep learning and Transformer frameworks achieve outstanding accuracy results in forecasting task, their emphasis lies in the exclusive decomposition of trend and seasonality factors from the time series or modeling the temporal dependencies. However, these frameworks rarely take exogenous features or external factors into account or simply use simple fully connected layers. As discussed above, we define external factors as all known variables that are not part of the target sequences to be forecasted. These external factors can be collected and utilized as additional dimensions alongside the target sequences for training purposes. In future applications, they may be available in advance to improve the forecasting accuracy of the target sequences. In the scope of our research, we exclude unknown external factors that may arise unexpectedly and have no precedent in historical data. DeepGLO [51] adapts the global-local combination mechanism to propose a hybrid model through a temporal convolution network (TCN) to combine the global prediction from a matrix factorization model regularized by a TCN with local raw time series and external factors. Nonetheless,

DeepGLO has neglected the incorporation of external factors concerning both the global context and the variability of their impact across distinct time series and temporal intervals. Temporal Fusion Transformer (TFT) [46] proposes Variable selection networks to select relevant input variables at each time step. TimeXer [31] is a Transformer-based model that enhances time series forecasting by effectively integrating exogenous variables through patch-wise self-attention and variate-wise cross-attention mechanisms. However, existing models often lack mechanisms to dynamically balance global and local influences of exogenous variables and to effectively capture interactions between external factors and specific local temporal patterns. Furthermore, the adaptability of these models into diverse encoder-decoder architectures as modular, plug-and-play components remains uncertain, raising questions about their flexibility and efficacy in modeling external factors across various frameworks.

2.5. Multimodal integration in sales forecasting

Recent advancements in multimodal integration has increasingly emphasized its application on time series forecasting in wide ranges of applications, particularly in sales and demand forecasting across diverse sectors such as retail, e-commerce, supply chain, and finance. In the sales forecasting task, the existing methods often fall short in capturing the complex interplay of various influencing factors. [52] proposed the Multi-modal Transform-based Fusion Model (M2TFM), which integrates product images, textual descriptions, and sales data using a Transformer-based architecture. This approach demonstrated improved forecasting accuracy for new product sales by effectively capturing the relationships between different data modalities. Similarly, [53] introduced a multimodal neural network that combines historical sales data with external information such as news articles and Google Trends. Their model showed a significant improvement in forecasting accuracy, highlighting the importance of incorporating diverse data sources to capture real-world events affecting demand. [54] developed the Modality-aware Transformer, which integrates textual reports and numerical time series data to forecast financial trends. By employing feature-level attention mechanisms, the model effectively captures the relevance of each modality, leading to more accurate and interpretable forecasts. [55] provided a comprehensive survey on multimodal time series analysis, categorizing existing methods based on their fusion strategies and highlighting the challenges in aligning heterogeneous data sources. Their work underscores the necessity of developing models capable of handling the complexity and diversity inherent in multimodal data. These studies collectively demonstrate the growing trend and effectiveness of integrating multimodal data in time series forecasting. Despite recent advances, significant challenges persist in effectively modeling the dynamic interactions among multiple modalities and ensuring model scalability across diverse domains. Notably, there remains a lack of a generalized plug-and-play module that can be seamlessly integrated into standard encoder-decoder architectures to incorporate external information into historical data. To address these limitations, we propose ConEm, a modular and adaptable framework designed to capture dynamic influences from both internal time series patterns and external contextual factors. By leveraging global embedding layers, ConEm models the dynamic impact of external inputs on both intra-series and inter-series correlations. Furthermore, its plug-and-play design facilitates easy integration into a wide range of encoder-decoder forecasting models, enhancing the accuracy of short- and mid-term future predictions. This framework presents a scalable and practical solution for multimodal integration in time series forecasting tasks.

3. The proposed framework

In this section, we provide the problem formulation for time series forecasting, and describe our proposal contextual embedding

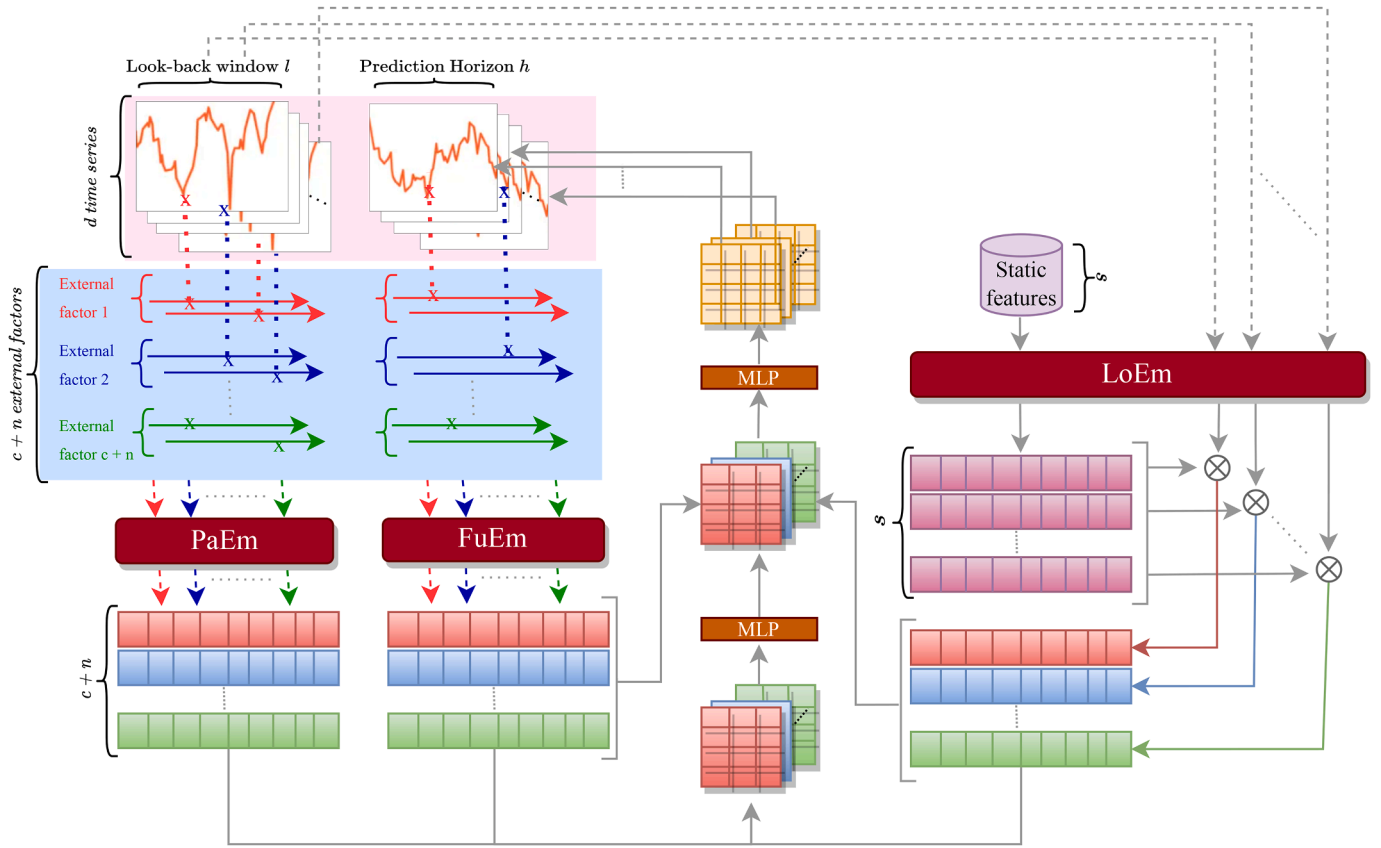


Fig. 3. The ConEm mechanism leverages internal and external factors to generate three parallel contextual embeddings—PaEm, FuEm, and LoEm—each capturing a distinct context. The feature data is enriched into a semantically meaningful representation. This combined representation is processed through MLPs to integrate information from the distinct contexts and transform it into a compact, unified form, which is subsequently incorporated into the context for each time point across multiple time series.

Table 1

The key mathematical notations.

Notation	Description
x^d	the target sequence
d	the number of target sequences
x^s	the static features
s	the number of static features
x_i^c	the time-varying categorical features
c	the number of time-varying categorical features
x_i^n	the time-varying numerical features
n	the number of time-varying numerical features
l	the look-back window size
h	the prediction horizon

mechanism for exogenous factors. This section also gives details how our proposed mechanism can be integrated with three well-known Transformer-based frameworks.

3.1. Problem formulation

Given the historical data of d number of time series at time point T : $H_T = [x_{T-l}^d, x_{T-l}^s, x_{T-l}^c, x_{T-l}^n] \dots [x_T^d, x_T^s, x_T^c, x_T^n]$ with the look-back window size of l , the task is to precisely forecast the target sequences x^d for each future time point: $H_{T+h} = [x_{T+1}^d \dots x_{T+h}^d]$, over the prediction horizon size of h , given $[x_{T+1}^c, x_{T+1}^n] \dots [x_{T+h}^c, x_{T+h}^n]$ as known future time-varying categorical and numerical features or external factors. x^s are all static features containing fixed information about all time series. We summarize the key notations in Table 1.

3.2. Contextual embedding mechanism

To mitigate the existing limitations mentioned in Sections 1 and 2, our proposed ConEm mechanism (Fig. 3) improves the forecasting accuracy for each time point based on the integration between local contexts and global relationships decomposed from both internal and external features. This mechanism is composed of three key components: Local Contextual Embedding (LoEm), Past Contextual Embedding (PaEm) and Future Contextual Embedding (FuEm). The intention is to construct a global latent representation for each external feature and to apply its global layer on target sequence based on specific local context. To construct the global layers, PaEm and FuEm help to model the global relationships between the external factors and target sequences from both past and future directions. The FuEm module is designed to capture future information by leveraging known future factors to identify and embed relevant correlation patterns. The PaEm module focuses on how past external factors can influence these future correlation patterns, which are essential for accurate forecasting. Because the impact of external factors varies on different target sequences, LoEm uses static features to adjust the global relationships between the external factors and each specific target sequence. In addition, the impact of external factors would also vary based on the time-varying factors. For example, the promotional activity increases (1) when recent unit sales are low, (2) in times of peak demand, and (3) after regular price changes [56]. Thus, LoEm also considers the input with the most recent values of target sequence as a time-varying factor to the external factors.

In the initial phase, all features are transformed by a specialized **Feature Tokenizer** block, taking inspired from [57], to reconstruct its

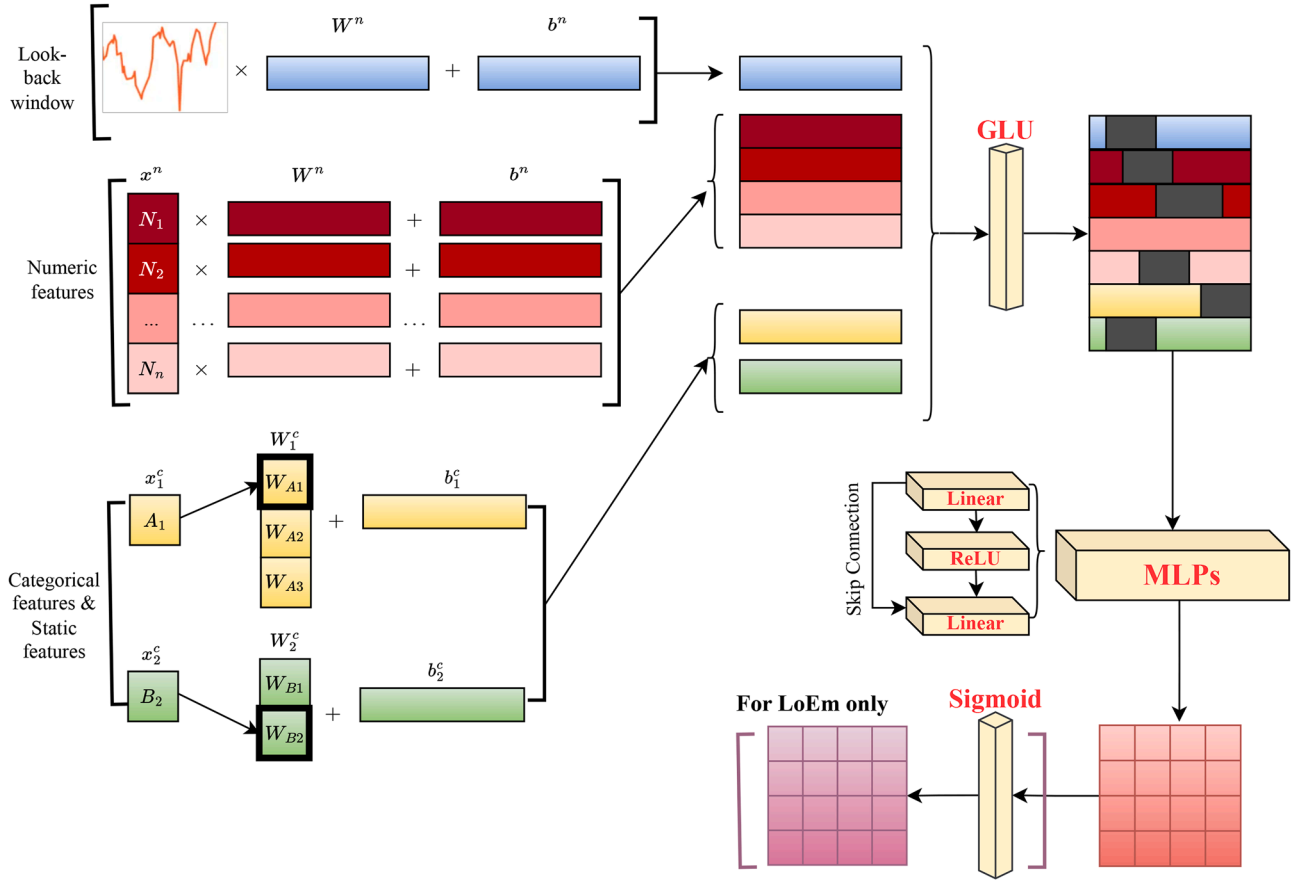


Fig. 4. The Feature Tokenizer converts look-back window time series, numerical features, categorical features, and static features into a more efficient semantic representation.

global latent representations (Fig. 4). As our aim is to identify correlations between target features and external features at each time point, enhancing forecasting accuracy for each individual time point. The task can be interpreted as tabular data structure for each time series at specific time point.

Our *Feature Tokenizer* block transforms numerical, categorical, and static features into a unified semantic space to enhance representation learning. Each feature x_j is mapped to a contextual embedding $C_j \in \mathbb{R}^d$ as $C_j = b_j + f_j(x_j)$, where $f_j(\cdot)$ applies a feature-specific transformation: numerical features use a learnable weight matrix W_n , while categorical features leverage a lookup table W_c and category-specific offsets. Specifically, for categorical features, embeddings are obtained using offset-indexed lookup: $x_i^{ce} = \text{Emb}_c(x_i^c + \text{category_offsets})$, where $\text{Emb}_c \in \mathbb{R}^{\sum c \times d_{\text{token}}}$ and $x_i^{ce} \in \mathbb{R}^{c \times d_{\text{token}}}$. Static categorical features are tokenized similarly: $x^{se} = \text{Emb}_s(x^s + \text{category_offsets})$, where $\text{Emb}_s \in \mathbb{R}^{\sum s \times d_{\text{token}}}$ and $x^{se} \in \mathbb{R}^{s \times d_{\text{token}}}$. This tokenized static features enable contextualized and individualized latent representations across diverse external feature types for each time series. Numerical features are linearly projected: $x_i^{nt} = W_n \cdot x_i^n$, where $W_n \in \mathbb{R}^{n \times n}$ and $x_i^n \in \mathbb{R}^{T \times n}$ for T time steps.

As ConEm aims to adjust the global layer based on each specific local context of each target sequence at a specific period, the look-back window sequence is also tokenized by Feature Tokenizer to be included in latent representation as LoEm. This helps contextual embeddings set up a local environment based the most recent time series data: $x_i^{dt} = W_d \cdot x_i^d$, where $W_d \in \mathbb{R}^{d \times d}$, $x_i^d \in \mathbb{R}^{T \times l}$, with T representing the look-back window size, and $x_i^{dt} \in \mathbb{R}^{T \times l}$ is the transformed x_i^d . After processing both numerical and categorical exogenous features, to share the correlations among time-varying feature embeddings, we concatenate them along the feature dimension for time-varying numerical and

categorical features to get its combination x_i^{ct} :

$$x_i^{ct} = \begin{cases} x_i^{nt} & \text{if } x_i^c \text{ is None} \\ [x_i^{nt}, x_i^{ce}] & \text{otherwise,} \end{cases} \quad (1)$$

Optional bias terms are added to the contextual embeddings x_i^{ct} , x^{se} , and x_i^{dt} , resulting in x_i^{cb} , x^{sb} , and x_i^{db} , respectively. Bias terms can help contextualize each embeddings independently, aligning them better in a shared latent space. These embeddings are then projected into the token space d_{token} using linear layers:

$$\begin{aligned} x_i^{co} &= W_o^c \cdot x_i^{cb}, \\ x^{so} &= W_o^s \cdot x^{sb}, \\ x_i^{do} &= W_o^d \cdot x_i^{db}, \end{aligned} \quad (2)$$

where $W_o^c, W_o^s, W_o^d \in \mathbb{R}^{d_{\text{bias}} \times d_{\text{token}}}$.

To retain salient information and suppress noise, each projection is passed through a Gated Linear Unit (GLU) [58]:

$$\begin{aligned} x_i^{cg} &= x_i^{co, \text{left}} \odot \sigma(x_i^{co, \text{right}}), \\ x^{sg} &= x^{so, \text{left}} \odot \sigma(x^{so, \text{right}}), \\ x_i^{dg} &= x_i^{do, \text{left}} \odot \sigma(x_i^{do, \text{right}}). \end{aligned} \quad (3)$$

Inspired by the efficiency of MLP-based models in time series forecasting [43–45], a shared MLP is applied to compress each embedding into a unified representation while still capturing temporal dependencies between exogenous features and target time series:

$$\begin{aligned} x_i^{cg} &\in \mathbb{R}^{t \times d_{\text{token}}} \rightarrow x_i^{tc} \in \mathbb{R}^{t \times d_{\text{model}}}, \\ x^{sg} &\in \mathbb{R}^{t \times d_{\text{token}}} \rightarrow x^{ts} \in \mathbb{R}^{t \times d_{\text{model}}}, \\ x_i^{dg} &\in \mathbb{R}^{t \times d_{\text{token}}} \rightarrow x_i^{td} \in \mathbb{R}^{t \times d_{\text{model}}}. \end{aligned} \quad (4)$$

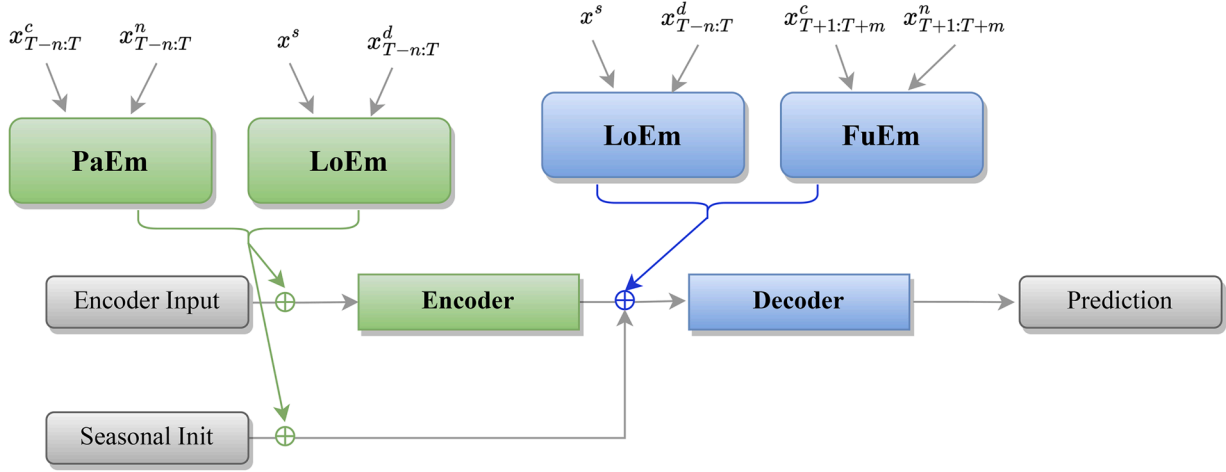


Fig. 5. The encoder-decoder pipeline incorporating our proposed ConEm mechanism.

The static feature outputs x^s are further processed by a sigmoid layer to produce relevance scores, which are used to weight the look-back embeddings. The final local contextualized representation x_t^{lc} is computed as $x_t^{lc} = x_t^s \odot x_t^{ld}$. Following tokenization by the Feature Tokenizer, the PaEm and FuEm embeddings are concatenated with LoEm to construct the Past and Future Contexts (Fig. 3). By incorporating LoEm, the model integrates static features and recent target patterns, enabling it to capture how the influence of external factors varies across time series. This fusion facilitates the modeling of inter-series correlations and dynamically adjusts the impact scale based on recent trends. As in the Feature Tokenizer, MLPs are applied to compress the concatenated representations to a compact form while preserving temporal and contextual semantics. The future context is further concatenated to emphasize the effect of external factors over the prediction horizon. In the next section, we describe how the proposed ConEm framework is incorporated into three representative Transformer-based models.

3.3. Encoder-decoder integration

This section illustrates how ConEm can be integrated seamlessly with three representative Transformer-based models: FEDformer [27], Autoformer [26], and Informer [25], following a consistent data processing pipeline (Fig. 5). The input sequence $x_{T-l:T}^d$ is first encoded via the Transformer's token embedding and Autoformer's series decomposition block [26] to extract temporal trends. Simultaneously, static features x^s are processed through the LoEm module to provide individualized contextual information.

For the future context, the initial trend component x_{aens} , derived from $x_{T-l:T}^d$ via series decomposition, is combined with x^s using LoEm. Time-varying categorical and numerical features from both the past and known future windows- $x_{T-l:T}^c$, $x_{T-l:T}^n$, $x_{T+1:T+h}^c$, $x_{T+1:T+h}^n$ -are tokenized via PaEm and FuEm, respectively. These are then modulated by LoEm to generate the final contextual embeddings for the encoder and decoder layers:

$$\mathcal{X}_{T-l:T}^{ct} = \text{PaEm}(x^d, x_{T-l:T}^c, x_{T-l:T}^n) \cdot \text{LoEm}(x^d, x_{T-l:T}^d), \quad (5)$$

$$x_{aens} = \text{SeriesDecomp}(x_{T-l:T}^d),$$

$$\mathcal{X}_{T+1:T+h}^{ct} = \text{FuEm}(x^d, x_{T+1:T+h}^c, x_{T+1:T+h}^n) \cdot \text{LoEm}(x^d, x_{aens}). \quad (6)$$

For the encoder input, the contextual embedding $\mathcal{X}_{T-l:T}^{ct}$ is added to the output of the Transformer's token embedding to enrich the representation with external factor information over the look-back window:

$$\mathcal{X}_{en} = \text{TokenEmbedding}(x_{T-l:T}) + \mathcal{X}_{T-l:T}^{ct}. \quad (7)$$

For the decoder input, the sequence $x_{T-l:T}^d$ is first processed by Autoformer's series decomposition block and the Transformer's token embedding to extract the seasonal component x_{aens} . To incorporate the

influence of external factors with potential seasonal patterns, the MLP-processed combination of Past and Future Contexts (Fig. 3) is added to x_{aens} before being passed to the decoder layers:

$$\mathcal{X}^{ct} = \text{MLP}(\text{Concat}([\mathcal{X}_{T-l:T}^{ct}, \mathcal{X}_{T+1:T+h}^{ct}])),$$

$$\mathcal{X}^{ct} = \text{MLP}(\text{Concat}([\mathcal{X}^{ct}, \mathcal{X}_{T+1:T+h}^{ct}])),$$

$$x_{aens} = \text{TokenEmbedding}(x_{aens}) + \mathcal{X}^{ct}. \quad (8)$$

When Informer is used as the backbone model, a similar data pipeline is applied for the encoder-decoder layers, with the exception of the LoEm input. Instead of relying on the initial trend component x_{aens} obtained from Autoformer's series decomposition, this configuration adopts the standard input format used in the vanilla Transformer.

In summary, the proposed ConEm framework is designed to be model-agnostic, enabling straightforward integration with various Transformer-based architectures. By enriching contextual embeddings with exogenous information from both static and temporal features, ConEm enhances the representational capacity of both encoder and decoder layers without requiring major modifications to the underlying backbone models.

3.4. Complexity analysis

In addition to the time complexity of the backbone models, the integration of our proposed ConEm module introduces an additional computational cost of $O(B \times (n + c) \times d_{\text{token}})$, where B denotes the batch size, n the number of time-varying features, c the number of static features, and d_{token} the dimensionality of the token embeddings. This added complexity primarily arises from the following sequential operations within our mechanism:

- Specialized feature tokenization: Each of the n time-varying and c static features is processed to generate initial embeddings.
- Contextual embedding generation: Three parallel contextual embeddings (past, future, and local) are constructed from the tokenized features.
- Integration with backbone models: The contextual embeddings are merged into the backbone model's architecture.

Each of these steps scales linearly with the total number of features and the embedding dimension d_{token} . Importantly, while the time complexity of backbone models typically scales with the input sequence length L -such as $O(L)$ for FEDformer, or $O(L \log L)$ for Autoformer and Informer-the complexity of ConEm is independent of L . This independence from sequence length makes ConEm particularly suitable for applications involving long time series or high-dimensional feature spaces.

Table 2

Data description of Australian pharmaceutical weekly dispensary.

No	Feature	Definition
1	Banner_group	Banner groups are groups of retail pharmacies similar to franchise groups.
2	Store_ID	ID of store containing product to be forecasted.
3	Master_item_ID	ID of product to be forecasted.
4	Bio_Equivalent_Code	Two pharmaceutical products are bioequivalent if they are pharmaceutically equivalent and their bioavailabilities (rate and extent of availability) after administration in the same molar dose are similar to such a degree that their effects, with respect to both efficacy and safety, can be expected to be essentially the same.
5	Molecule	The molecule information of product.
6	Patent_Group	Products are classified into three categorical values: Branded, Substitutable and Generic. Branded products are protected by patent and have no substitutes. Substitutable products are original products that now have generic substitutes. Generic products considered bioequivalent to Substitutable products.
7	Week_of_Month	Week in month.
8	Week_of_Year	Week in year.
9	Month_of_Year	Month in year.
10	Day_of_Safety_Net	Last date of calendar year as PBS Safety Net usually resets at the beginning of each year, 1 January.
11	Holiday	If the date is a public holiday date.
12	City	City of store.
13	State	State of store.
14	Postcode	Postcode of store.
15	Price_change	The percentage price difference comparing to the previous week.

Table 3

Data description of Australian daily POS.

No	Feature	Definition
1	Banner_group	Banner groups are groups of retail pharmacies similar to franchise groups.
2	Store_ID	ID of store containing product to be forecasted.
3	Master_item_ID	ID of product to be forecasted.
4	Day_of_Week	Day in week.
5	Day_of_Month	Day in month.
6	Day_of_Year	Day in year.
7	Month_of_Year	Month in year.
8	Day_of_Safety_Net	Last date of calendar year as PBS Safety Net usually resets at the beginning of each year, 1 January.
9	Holiday	If the date is a public holiday date.
10	Promo_date	If the date is in a promotion campaign.
11	Price_to_previous	The percentage price difference comparing to the latest previous 7-day period without any promotion campaign.
12	Price_to_previous_with-out_promo	The percentage price difference comparing to the latest previous 7-day period which may be with or without promotion.
13	Promo_length	The length of promotion campaign as number of days.
14	City	City of store.
15	State	State of store.
16	Postcode	Postcode of store.

4. Experiments

4.1. Experimental data

We conducted extensive forecasting experiments on three real-world datasets from retail domain: Australian Pharmaceutical Weekly Dispensary,

Table 4

Data description of corporacion favorita grocery sales.

No	Feature	Definition
1	Store_Number	ID of store containing product to be forecasted.
2	Item_Number	ID of product to be forecasted.
3	Day_of_Week	Day in week.
4	Day_of_Month	Day in month.
5	Day_of_Year	Day in year.
6	Month_of_Year	Month in year.
7	onpromotion	Binary feature if item is in promotion.
8	is_national_holiday	Binary feature if the date is National holiday date.
9	is_holiday_state	Binary feature if the date is State holiday date.
10	is_holiday_city	Binary feature if the date is City holiday date.
11	is_payday	Binary feature if the date is public sector paid date. Wages in the public sector are paid every two weeks on the 15 th and on the last day of the month. Supermarket sales could be affected by this.
12	perishable	Binary feature if item is perishable.
13	city	City of store.
14	state	State of store.
15	family	Primary category of product.
16	class	Primary class of product.

sary, Australian Daily POS and Corporacion Favorita Grocery Sales². In addition, the model was evaluated on different domains for generalization evaluation. We utilized Weather³ dataset for long-term forecasting. For short-term forecasting tasks, we incorporate the short-term electricity price forecasting (EPF) datasets [59], which represent real-world scenarios involving exogenous variables.

4.1.1. Australian pharmaceutical weekly dispensary

Dataset includes weekly sales data of 8 stores and top 2000 prescription-only medicines, that can be either chronic (ongoing), acute (short-term), or lifestyle (oral contraceptives, weight loss etc.), from January 2018 to August 2022. As prescription-only medicines are slow-moving products with low daily demand, we aggregate the daily data to weekly data and set weekly forecasting settings. A summary of the data description for the Australian Pharmaceutical Weekly Dispensary is provided in Table 2. Static features keep an important role as each medicine has its own specialized characteristics such as molecule information. We split data from January 2018 to December 2021 for training data and 2022 for testing data.

4.1.2. Australian daily POS

Dataset includes daily sales data of 398 pharmaceutical stores and top 100 fast-moving items, which are commonly found in grocery, such as vitamins, hair care, beauty and baby categories. These items rely on promotional activity and consumer preference to drive demand. Regarding the promotion information, a binary feature is used to check if the date is in promotion period. As there would be consecutive promotion periods with different discount rates, there are two numeric features measuring the percentage price difference comparing to the previous period without promotion and the percentage price difference comparing to the latest previous period with or without promotion. Table 3 provides a summary of the data description for the Australian Pharmaceutical Weekly Dispensary. We split data from March 2021 to May 2022 for training data and from June 2022 to October 2022 for testing data.

4.1.3. Corporación favorita grocery sales

Dataset has been used in the 2017 Kaggle competition and many research studies [9,16]. Corporación Favorita is an Ecuadorian company owner of multiple supermarkets across Latin America, releasing this daily sales data for 4400 unique items in 54 different Ecuadorian stores from January 1st 2013 to August 15th 2017. Table 4 presents a

² <https://www.kaggle.com/c/favorita-grocery-sales-forecasting>

³ <https://www.bgc-jena.mpg.de/wetter/>

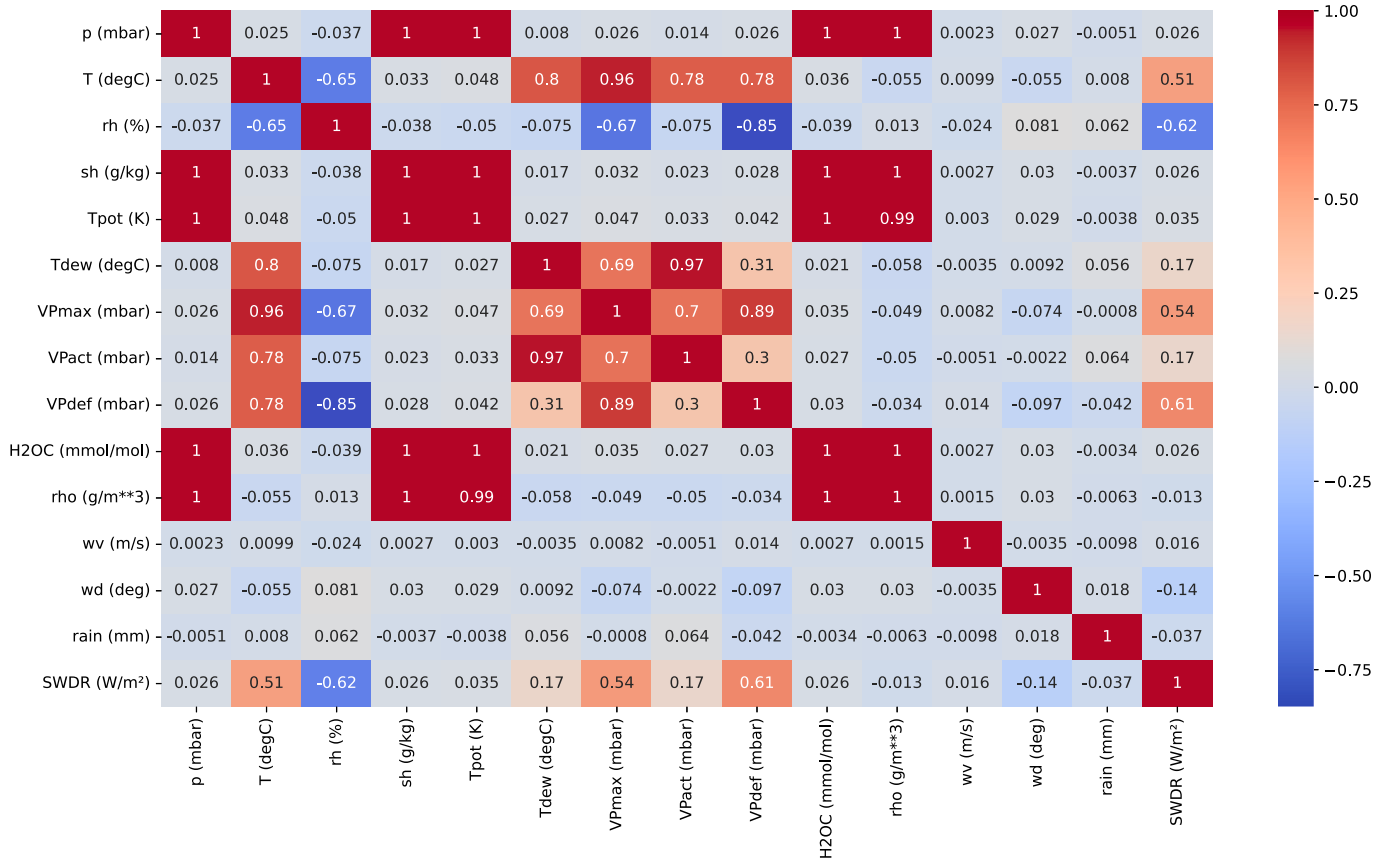


Fig. 6. Correlation matrix of weather measurements.

summary of the data description for the Corporación Favorita Grocery Sales. Data is split from January 1st 2013 to December 31st 2016 for training data and from January 1st 2017 to August 15th 2017 for testing data.

4.1.4. Weather

Dataset is collected from two weather stations in Germany : Saaleaue and Beutenberg, for every 10-minute period which is filtered from 2022-01-01 0:00 to 2023-04-02 6:00 for our experiments. Only one static feature (Station Location: Saaleaue or Beutenberg) is used. There are 15 time series for different weather measurements included in Weather data: p (mbar), T (degC), rh (%), sh (g/kg), $Tpot$ (K), $Tdew$ (degC), $VPmax$ (mbar), $VPact$ (mbar), $VPdef$ (mbar), $H2OC$ (mmol/mol), ρ (g/m**3), ww (m/s), wd (deg), $rain$ (mm), and $SWDR$ (W/m). The main purpose of this experimental setting on Weather dataset is to assess the model capability in modeling the impact from external factors on a different domain through various highly-correlated weather measurements across different time series as seen in Fig. 6. The training data is from 2022-01-01 0:00 to 2023-01-01 0:00. The test data is from 2023-01-01 0:10 to 2023-04-02 6:00.

4.1.5. EPF

These datasets are constructed from six years of historical data across five major power markets (NP, PJM, BE, FR, DE). Each dataset includes electricity price as the target endogenous variable, along with two practically significant exogenous variables known to influence price dynamics. EPF consists of five subsets, each representing a different day-ahead electricity market, with six years of historical data. The subsets are described as follows:

- NP (Nord Pool): Hourly electricity prices, grid load, and wind power forecasts from Jan 1, 2013 to Dec 24, 2018. It includes two exogenous features: Grid Load, Wind Power.

- PJM: Covers the Pennsylvania-New Jersey-Maryland market with zonal electricity prices (COMED), system load, and COMED load forecasts (same period). It includes two exogenous features: System Load, SyZonal COMED load.
- BE (Belgium): Includes hourly prices, Belgian load forecasts, and French generation forecasts from Jan 9, 2011 to Dec 31, 2016. It includes two exogenous features: Generation, System Load.
- FR (France): Provides hourly prices, load, and generation forecasts from Jan 9, 2012 to Dec 31, 2017. It includes two exogenous features: Generation, System Load.
- DE (Germany): Contains hourly prices, zonal load forecasts (Amprion TSO), and wind/solar generation forecasts from Jan 9, 2012 to Dec 31, 2017. It includes two exogenous features: Wind power, Ampirion zonal load.

4.2. Implementation details

Regarding sales and weather dataset, we conducted evaluation experiments for our proposal ConEm mechanism integrated with three Transformer variants: Autoformer [26], Informer [25] and FEDformer [27]. The results were also benchmarked against five additional models: N-BEATS [41], TFT [46], Pyraformer [47], TimesNet [37] and N-HiTS [42]. The internal and external factors were embedded into the baseline models by its original pipelines if available. If the model lacks an embedding mechanism for internal and external factors, it was trained and evaluated exclusively on the time series data. A grid search strategy was employed for our proposed model and all baseline models to identify the optimal hyper-parameter settings. We did not apply various enhanced practices for deep learning model such as pretraining, data augmentation, learning rate decay, etc. Regarding N-BEATS, TFT and N-HiTS, the widely-used time series forecasting package PyTorch

Table 5

Comparison of forecasting errors between the backbone models without and with ConEm. The analysis on the two datasets: Australian Pharmaceutical Weekly Dispensary and Australian Daily POS, is conducted by increasing the look-back window and prediction horizon sizes.

Dataset	Look-back window	Prediction horizon	Autoformer[26]		+ ConEm		Informer [25]		+ ConEm		FEDformer [27]		+ ConEm	
			MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓
Australian Pharmaceutical Weekly Dispensary	$I = 4$	$O = 4$	47.50	2.77	43.36	2.62	112.55	3.45	103.31	3.34	45.88	2.71	38.06	2.42
		$O = 6$	42.86	2.89	38.41	2.64	110.05	3.44	101.22	3.28	45.27	2.84	35.51	2.42
		$O = 8$	43.22	2.88	39.78	2.63	108.81	3.46	100.61	3.29	44.50	2.82	34.78	2.43
	$I = 8$	$O = 4$	53.17	2.86	45.21	2.72	104.27	3.49	93.40	3.36	50.64	2.77	39.27	2.39
		$O = 8$	44.02	2.71	36.42	2.53	113.57	3.51	94.32	3.38	43.76	2.73	37.46	2.41
		$O = 12$	44.96	2.73	37.35	2.56	117.26	3.55	109.23	3.42	44.79	2.73	38.45	2.46
Australian Daily POS	$I = 15$	$O = 15$	3.48	1.19	3.11	1.13	3.43	1.21	3.08	1.14	3.49	1.18	3.11	1.11
		$O = 21$	3.48	1.18	3.14	1.11	3.46	1.18	3.12	1.13	3.54	1.18	3.15	1.10
		$O = 30$	3.55	1.19	3.19	1.15	3.48	1.18	3.18	1.13	3.51	1.17	3.18	1.12
	$I = 30$	$O = 30$	3.24	1.17	2.86	1.13	3.10	1.17	2.77	1.12	3.22	1.17	2.84	1.12
		$O = 60$	3.50	1.22	3.09	1.15	3.20	1.18	2.89	1.12	3.32	1.18	2.92	1.13
		$O = 90$	4.27	1.30	3.74	1.21	3.69	1.20	3.24	1.14	3.86	1.23	3.40	1.16

Table 6

Prediction performance comparisons on Australian Pharmaceutical Weekly Dispensary. Best results are in **bold**. Second best results are underlined.

Look-back window	$I = 4$						$I = 8$					
	$O = 4$		$O = 6$		$O = 8$		$O = 4$		$O = 8$		$O = 12$	
Prediction horizon												
Model	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓
N-BEATS [41]	44.78	2.59	47.23	2.74	44.47	2.73	43.37	<u>2.41</u>	43.55	2.52	48.15	2.65
Autoformer [26]	47.50	2.77	42.86	2.89	43.22	2.88	53.17	2.86	44.02	2.71	44.96	2.73
Informer [25]	112.55	3.45	110.05	3.44	108.81	3.46	104.27	3.49	113.57	3.51	117.26	3.55
TFT [46]	120.49	3.65	115.71	3.64	120.87	3.71	113.65	3.58	111.77	3.59	137.41	3.79
Pyraformer [47]	52.29	2.87	50.13	2.81	51.87	2.81	59.85	2.82	56.73	2.78	53.51	2.79
FEDformer [27]	45.88	2.71	45.27	2.84	44.50	2.82	50.64	2.77	43.76	2.73	44.79	2.73
N-HiTS [42]	41.25	2.47	41.86	2.52	39.49	2.50	40.37	2.48	37.59	2.44	43.75	2.68
TimesNet [37]	42.32	2.53	40.86	<u>2.49</u>	<u>38.74</u>	<u>2.49</u>	41.54	2.49	38.34	2.48	<u>38.21</u>	<u>2.55</u>
<i>Ours - Autoformer-based</i>	43.36	2.62	38.41	2.64	39.78	2.63	45.21	2.72	36.42	2.53	37.35	2.56
<i>Ours - Informer-based</i>	103.31	3.34	101.22	3.28	100.61	3.29	93.40	3.36	94.32	3.38	109.23	3.42
<i>Ours - FEDformer-based</i>	38.06	2.42	35.51	2.42	34.78	2.43	39.27	2.39	<u>37.46</u>	2.41	38.45	2.46

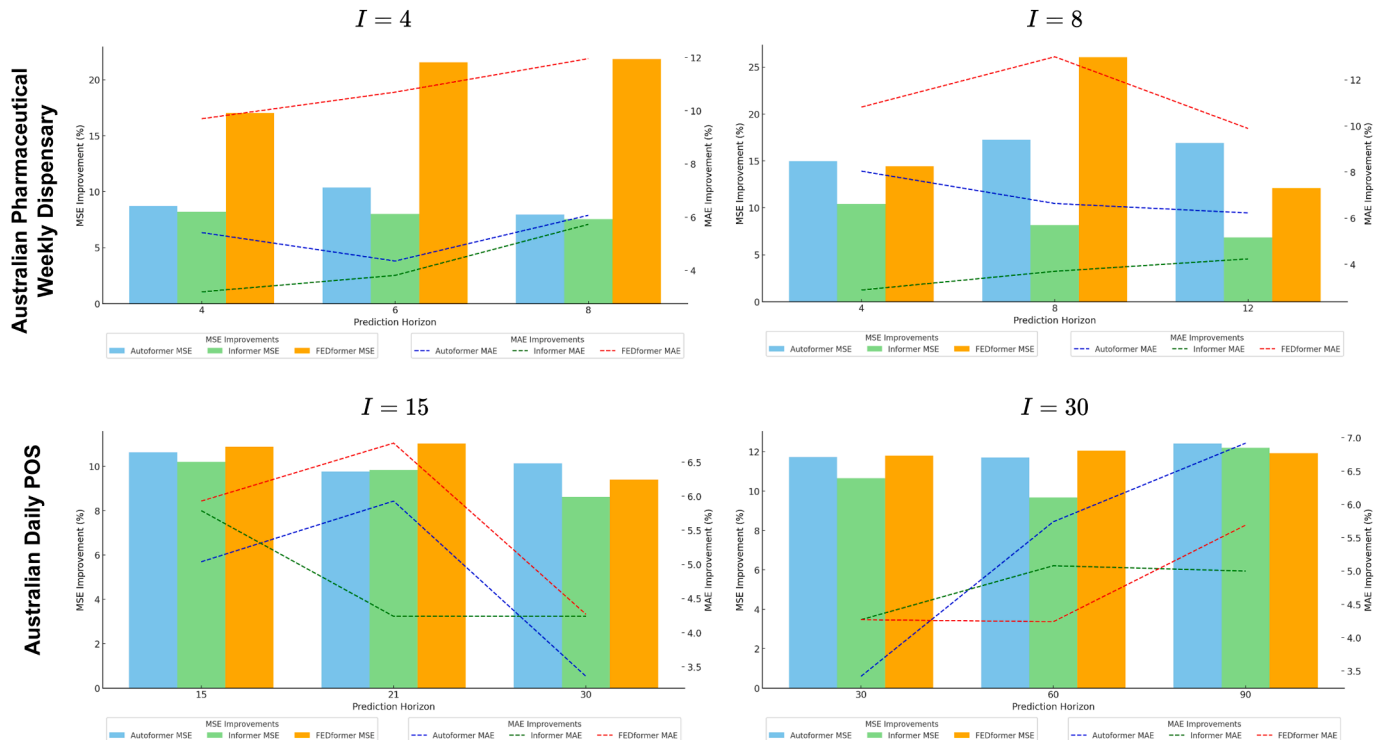


Fig. 7. Percentage improvement of backbone models with ConEm across two datasets: Australian Pharmaceutical Weekly Dispensary and Australian Daily POS. The analysis examines the impact on the increasing look-back window and prediction horizon sizes.

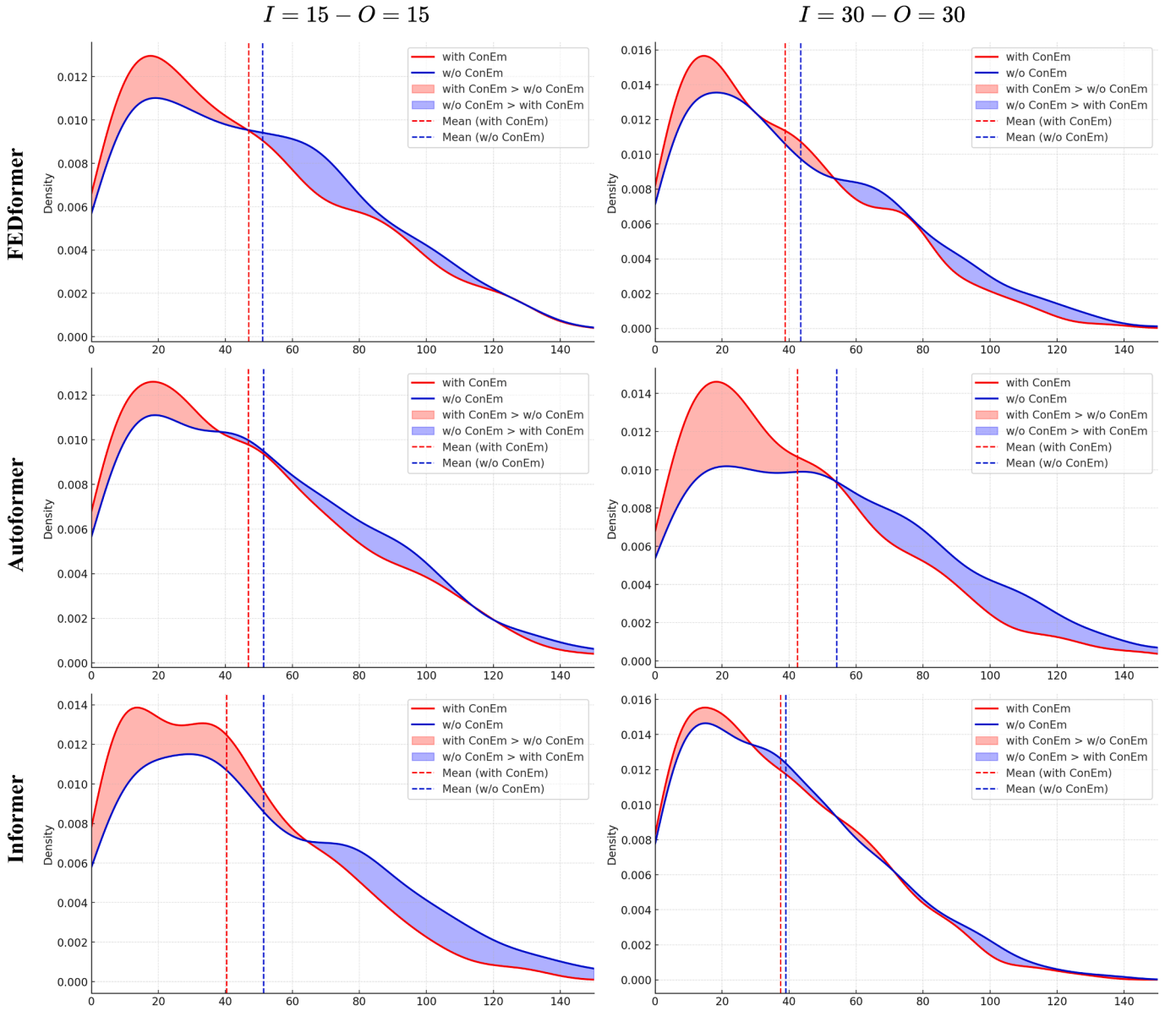


Fig. 8. Density distributions of MAE for Australian daily POS of the top 10 high-demand, highly-promoted products across the top 100 sales stores, analyzed under $I=15-O=15$ and $I=30-O=30$ settings. MAE values are scaled up by a factor of 14 to represent a two-week period. The blue line represents the backbone models without our ConEm. The red line represents the backbone models with our ConEm. The dash lines indicates the means of each distribution.

Forecasting was implemented with its hyper-parameter tuning function for 50 trials and 50 epochs to find the best hyper-parameters for each model, then the models with the best hyper-parameters were trained for 200 epochs. Regarding short-term electricity price forecasting - EPF dataset, we evaluated our ConEm mechanism with five backbone models: Autoformer [26], Informer [25], FEDformer [27], PatchTST [48] and iTransformer [49]. As encoder-only architectures, we evaluated two integration strategies with PatchTST [48] and iTransformer [49]: (i) incorporating our mechanism within the encoder itself, and (ii) appending it to the encoder output prior to the linear forecasting layer. We adopt the standardized experimental protocol of NBEATSx [60] and TimeXer [31], setting the input sequence length to 168 and the prediction horizon to 24 time steps. For comparative purposes, we also reused the results reported in TimeXer [31] for evaluating the performance of the respective frameworks: TimeXer [31], NBEATSx [60], TFT [46], Stationary [61], RLinear [50], Crossformer [28], TiDE [62], TimesNet [37], DLinear [44], and SCINet [36]. All experimental results were evalu-

ated by two primary metrics: MAE and MSE. Our experiments were conducted based on the design of existing frameworks [25–27,31,60], which guided our definition of varying sizes of look-back window I and prediction horizon O :

- Australian Pharmaceutical Weekly Dispensary: $I \in \{4, 8\}$ and $O \in \{4, 6, 8, 12\}$ (weeks).
- Australian Daily POS: $I \in \{15, 30\}$ and $O \in \{15, 21, 30, 60, 90\}$ (days).
- Corporacion Favorita Grocery Sales: $I \in \{32\}$ and $O \in \{16, 24, 32\}$ (days).
- Weather: $I \in \{96\}$ and $O \in \{96\}$ (10-minute periods).
- EPF: $I \in \{168\}$ and $O \in \{24\}$ (1-hour periods).

Regarding sales forecasting experiments, the learning rate was set as 1×10^{-4} and dropout rate was set as 1×10^{-2} . All models were fine-tuned with one or two layers for the encoder and decoder modules. The Wavelet version was used for both our proposal and FEDformer. TimesNet was fine-tuned by 5–10 for top_k parameter and 3–6 for number

Table 7Prediction performance comparisons on Australian Daily POS. Best results are in **bold**. Second best results are underlined.

Look-back window	$I = 15$						$I = 30$					
	$O = 15$		$O = 21$		$O = 30$		$O = 30$		$O = 60$		$O = 90$	
Prediction horizon												
Model	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓
N-BEATS [41]	5.85	1.29	4.34	1.23	4.27	1.16	5.28	1.27	4.27	1.22	4.25	1.21
Autoformer [26]	3.48	1.19	3.48	1.18	3.55	1.19	3.24	1.17	3.50	1.22	4.27	1.30
Informer [25]	3.43	1.21	3.46	1.18	3.48	1.18	3.10	1.17	3.20	1.18	3.69	1.20
TFT [46]	3.31	1.20	3.33	1.29	3.29	1.28	3.03	1.19	3.08	1.20	3.53	1.23
Pyraformer [47]	3.54	1.23	3.43	1.19	3.38	1.18	3.12	1.18	3.09	1.17	3.13	1.20
FEDformer [27]	3.49	1.18	3.54	1.18	3.51	1.17	3.22	1.17	3.32	1.18	3.86	1.23
N-HITS [42]	5.56	1.27	5.53	1.27	5.48	1.27	5.24	1.27	5.29	1.27	5.55	1.27
TimesNet [37]	3.89	1.24	3.41	1.21	3.27	1.19	3.24	1.17	3.29	1.18	3.25	1.17
<i>Ours - Autoformer-based</i>	<u>3.11</u>	<u>1.13</u>	<u>3.14</u>	<u>1.11</u>	<u>3.19</u>	1.15	2.86	1.13	3.09	1.15	3.74	1.21
<i>Ours - Informer-based</i>	3.08	1.14	3.12	1.13	3.18	<u>1.13</u>	2.77	1.12	2.89	1.12	<u>3.24</u>	1.14
<i>Ours - FEDformer-based</i>	<u>3.11</u>	1.11	3.15	1.10	3.18	1.12	<u>2.84</u>	1.12	<u>2.92</u>	<u>1.13</u>	3.40	<u>1.16</u>

of kernels parameter. Regarding weather forecasting experiments, we conducted simulation experiments for nine weather measurements: T (degC), rh (%), $Tdew$ (degC), $VPmax$ (mbar), $VPact$ (mbar), $VPdef$ (mbar), wv (m/s), wd (deg), and $rain$ (mm). There are additional five temporal features: hour of day (with 24 distinct values), day of week (with 7 distinct values), day of month (with 31 distinct values), day of year (with 366 distinct values), and month of year (with 12 distinct values). Time series: p (mbar), sh (g/kg), $Tpot$ (K), $H2OC$ (mmol/mol), and ρ (g/m³), were not used for target time series in our experiments as they contain missing values for 2022-01-21 22:00 to 2022-01-23 18:30 period, whose values are assigned as -9999 in raw data. Time series $SWDR$ (W/m) was not used for target time series as it contains nearly 50% of values as zero value. Thus, these six weather time series were used as external factors for our simulation experiments. The learning and dropout rates are both set as 1×10^{-5} . The numbers of encoder-decoder modules are fine-tuned as one or two layers. The Wavelet version is used for both our proposal and FEDformer. TimesNet uses 5 for top_k parameter and 6 for number of kernels parameter. A 0.8-0.2 train-validation split ratio was applied across all training experiments. Regarding EPF dataset, we followed the baselines [31,60] with train-valid-test sizes as (36500, 5219, 10460). We conducted five experiments for each model and used the best result for comparison.

4.3. Main results

4.3.1. Australian pharmaceutical weekly dispensary

Our proposal demonstrates improvements across all forecasting scenarios compared to backbone models, achieving enhancements of 8% to 26% in MSE and 3% to 15% in MAE (Table 5 and Fig. 7). Specifically, compared to the FEDformer backbone, the proposed ConEm achieves an average relative MSE reduction of 20% and 18% for the two input settings, respectively, along with an average 11% reduction in MAE. ConEm enables Autoformer and Informer to achieve average reductions of 9% in MSE and 5% in MAE for a 4-week look-back window and 12% in MSE and 5% in MAE for an 8-week look-back window. The backbone models integrated with our ConEm mechanism also achieve SOTA results compared to other baseline models, especially FEDformer-based variant (Table 6). FEDformer with ConEm achieves SOTA results in both MAE and MSE in four settings while it also achieves best MAE results in $I=8-O=8$ and $I=8-O=12$ settings. Autoformer with ConEm attains the best MSE results in the $I=8-O=8$ and $I=8-O=12$ settings.

4.3.2. Australian daily POS

There are also significant improvements for all backbone models with ConEm mechanism in all forecasting settings (Table 5 and Fig. 7). To more effectively evaluate the impact of our mechanism on backbone models through the integration of external factors, we selected

the top 10 high-demand and highly-promoted products-those associated with the greatest number of promotional campaigns-across the top 100 highest-performing sales stores. The reason is that daily sales for all items tend to be right-skewed, as many items have low or zero sales on a given day. Additionally, promotions are often concentrated on certain categories or products, such as vitamins. We then scaled up the MAE by a factor of 14 to represent a two-week period, making it easier to observe the impact through larger values (Fig. 8). Density distribution plots make it easier to observe that our MAE distributions tend to be higher to the left, with lower distribution means compared to the results from the backbone models without our mechanism. These results demonstrate the effectiveness of our mechanism in capturing the influence of external factors, thereby reducing the MAE in time series forecasting for each backbone model. A notable observation from Fig. 7 is the increasing percentage improvement for longer prediction horizon sizes when using a look-back window size of 30 days in MAE. However, the percentage improvement decreases for longer prediction horizon sizes for the look-back window size of 15 days. Compared to other baseline methods, it is somewhat surprising that ConEm, when utilizing Informer as its backbone model, achieves competitive SOTA performance across five prediction settings in terms of MSE and demonstrates superior performance in all settings with a 30-day look-back window in terms of MAE. FEDformer with ConEm achieves SOTA results in almost all settings in MAE except $I=30-O=60$ and $I=30-O=90$ settings (Table 7).

4.3.3. Corporacion favorita grocery sales

Compared to the backbone models, our proposed approach demonstrates consistent improvements across all forecasting scenarios, achieving notable reductions in MAE ranging from 8% to 13% for FEDformer, 2% to 7% for Autoformer, and 4% to 9% for Informer (Table 8). In terms of MSE, a clear pattern emerges where our mechanism consistently enhances performance as the prediction horizon size increases across most scenarios: 1.78% $\rightarrow$ 2.85% $\rightarrow$ 3.76% ($O=16 \rightarrow O=24 \rightarrow O=32$) for Autoformer, 4.96% $\rightarrow$ 5.90% $\rightarrow$ 7.73% ($O=16 \rightarrow O=24 \rightarrow O=32$) for FEDformer, and 6.90% $\rightarrow$ 7.74% ($O=16 \rightarrow O=24$) for Informer. Similar to the Australian Daily POS dataset, Informer integrated with our ConEm mechanism achieves SOTA performance across all scenarios in both MSE and MAE, outperforming the second-best models by 3.5% to 4% in the $O=16$ and $O=24$ settings (Table 9).

4.3.4. Weather

In term of long-term forecasting settings, all three backbone models have seen an improvement in accuracy through the integration with our proposed contextual embedding mechanism (Table 10). As seen in qualitative analysis examples of T (degC) and rh (%) (Fig. 9 and Fig. 10), FEDformer, Autoformer and Informer have seen a significantly improved accuracy by our proposal contextual mechanism



Fig. 9. Prediction case of Transformer-based models with and without our ConEm mechanism on T ($degC$) of Weather benchmark under the I -96- O -96 setting.



Fig. 10. Prediction case of Transformer-based models with and without our ConEm mechanism on rh (%) of Weather benchmark under the I -96- O -96 setting.

through modeling the impact from the external factors on T ($degC$) and rh (%), resulting in significant improvements in capturing the local time point. FEDformer with our proposal ConEm also outperforms other baselines with the SOTA results for six out of nine evaluation measurements (Tables 11–13). It can be observed that there are significant improvements in six measurements: T ($degC$), rh (%), $Tdew$ ($degC$), $VPmax$ (mbar), $VPact$ (mbar), and $VPdef$ (mbar), because these time series are highly-correlated to many another time series (Fig. 6), which are assumed to be known in advance and used as external factors in our simulation. As a result, our model significantly enhances the time series forecasting by effectively incorporating external factors. This leads

to a notable improvement in performance compared to the second-best models. Demonstrating the model's ability to capture the impact of external factors on the target time series fulfills our primary objective in this simulation setting.

4.3.5. EPF

Our quantitative results across five regional datasets demonstrate consistent improvements in forecasting accuracy when integrating the proposed ConEm mechanism (Tables 14 and 15). On average, ConEm reduces MSE/MAE by 19.5%/13.2% for Autoformer, 81.0%/69.3% for Informer, and 26.8%/17.1% for FEDformer. For PatchTS, the +ConEm

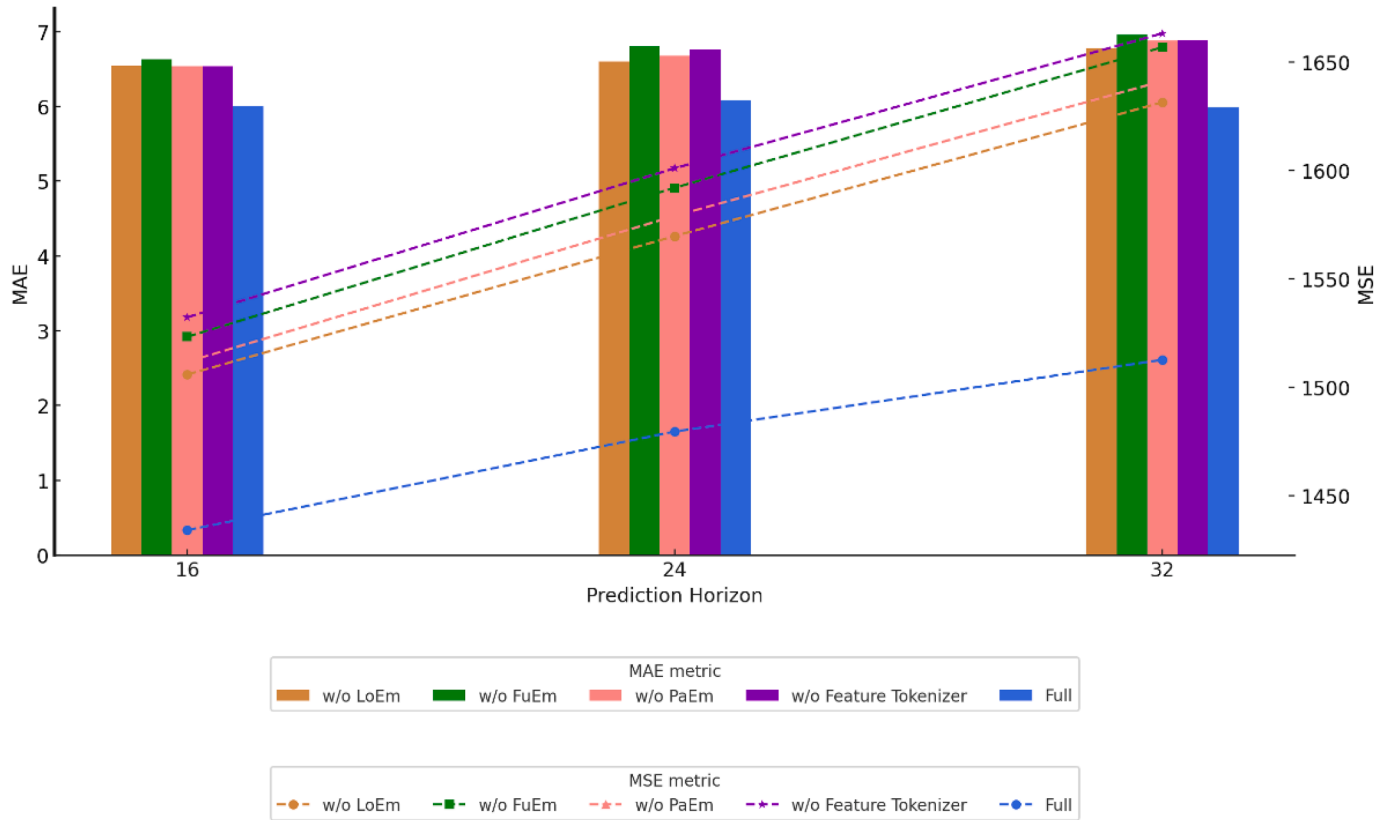


Fig. 11. Ablation studies for short-term forecasting on the Corporación Favorita Grocery Sales dataset, using a look-back window size I of {32} (days) and the prediction horizon sizes O of {16,24,32} (days).

Table 8

Comparison of short-term forecasting performance. The experiment is conducted on Corporacion Favorita Grocery Sales. We analyze the forecasting error of the backbone models with and without ConEm by increasing the prediction horizon from 16 to 32 while the look-back window is fixed to 32.

Prediction horizon	$O = 16$		$O = 24$		$O = 32$	
Metric	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓
Autoformer [26]	6.6119	1495.6	6.8435	1571.0	6.8237	1639.1
+ ConEm	6.2227	1468.7	6.4914	1526.3	6.6985	1630.1
Informer [25]	5.7138	1468.4	5.9213	1482.1	6.3101	1548.2
+ ConEm	5.4862	1367.1	5.6865	1428.4	5.7743	1484.1
FEDformer [27]	6.5692	1508.9	6.7845	1572.2	6.8923	1639.3
+ ConEm	5.9983	1434.0	6.0840	1479.5	5.9865	1512.6

Table 9

Prediction performance comparisons on Corporacion Favorita Grocery Sales. Best results are in **bold**. Second best results are underlined.

Look-back window	$I = 32$					
Prediction horizon	$O = 16$		$O = 24$		$O = 32$	
Model	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓
N-BEATS [41]	11.242	1543.3	11.313	1584.2	11.224	1605.1
Autoformer [26]	6.6119	1495.6	6.8435	1571.0	6.8237	1639.1
Informer [25]	5.7138	1468.4	5.9213	1482.1	6.3101	1548.2
TFT [46]	8.6282	1425.5	7.8409	1420.6	8.1082	1586.6
Pyraformer [47]	<u>5.6822</u>	<u>1423.5</u>	5.9323	1483.4	6.3321	1568.3
FEDformer [27]	6.5692	1508.9	6.7845	1572.2	6.8923	1639.3
N-HiTS [42]	10.823	1459.3	10.897	1496.2	10.880	1542.1
TimesNet [37]	5.9234	1445.7	<u>5.9118</u>	<u>1480.0</u>	6.2812	1587.8
Ours - Autoformer-based	6.2227	1468.7	6.4914	1526.3	6.6985	1630.1
Ours - Informer-based	5.4862	1367.1	5.6865	1428.4	5.7743	1484.1
Ours - FEDformer-based	5.9983	1434.0	6.0840	1479.5	<u>5.9865</u>	<u>1512.6</u>

variant achieves a 17.7 % MSE and 14.4 % MAE reduction compared to the baseline, while for iTransformer, it yields 8.7 % and 6.7 % improvements, respectively. Across all models, appending ConEm after the encoder (+ ConEm) consistently outperforms embedding it within the encoder (+ ConEm (E)). Table 16 provides a comprehensive comparison of 15 forecasting models on the EPF dataset. Among all methods, Informer + ConEm achieves the best overall performance with the lowest average MSE (0.277) and MAE (0.245), followed closely by PatchTST + ConEm (MSE: 0.280, MAE: 0.255). PatchTST + ConEm achieves the lowest MSE on the PJM dataset (0.099) and second-best performance across NP, BE, DE, and overall average, demonstrating robust generalization. Notably, all ConEm-enhanced variants outperform their original backbones in both MSE and MAE on average. These results validate ConEm as an effective plug-and-play enhancement that can consistently improve forecasting models across diverse datasets and architectures.

4.4. Ablation studies

4.4.1. Embedding contributions

We conducted the ablation studies with FEDformer-based backbone model and three variants without FuEm, LoEm, or PaEm components for all three short-term forecasting target sequence lengths O of {16,24,32} (days) (Fig. 11), and select T (degC) and rh (%) for long-term forecasting experiments (Table 17). In the scenario without the Feature Tokenizer, the baseline is established by applying a single linear layer transformation to the exogenous features, which are then added to the tokenized target values. As seen in short-term forecasting experiments (Fig. 11), FuEm keeps the most important role among three components as models without FuEm have the most significant decrease in forecasting accuracy ranging from 10 % to 16 % for MAE and from 6 % to 10 % for MSE. It is similar for the long-term forecasting settings (Table 17) as FuEm continues to maintain its critical role, demonstrating the most

Table 10

Comparison of forecasting errors between the backbone models without and with ConEm. The analysis on the Weather dataset, is conducted by the fixed look-back window and fixed prediction horizon sizes.

Weather measurement	Autoformer [26]		+ ConEm		Informer [25]		+ ConEm		FEDformer [27]		+ ConEm	
	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓
rain (mm)	0.0233	0.0044	0.0166	0.0043	0.0207	0.0045	0.0177	0.0044	0.0230	0.0043	0.0155	0.0042
rh (%)	5.7257	55.650	3.6424	13.267	4.7953	38.448	2.4865	10.540	4.4780	34.108	0.6565	0.7572
T (degC)	1.5169	3.9537	0.5932	0.6377	1.4631	3.6560	1.1545	2.3377	0.9804	1.5919	0.1341	0.0343
Tdew (degC)	1.2092	2.5228	0.8689	1.3016	1.1778	2.5829	0.5399	0.4970	0.7617	0.9600	0.0831	0.0142
VPmax (mbar)	1.5275	3.8926	1.1050	2.8171	2.1572	7.1303	1.9521	6.5402	1.0445	1.9000	0.1242	0.0282
VPdef (mbar)	1.1536	2.5306	1.0645	2.3658	1.2478	2.9590	0.5497	0.6393	0.8511	1.4185	0.2206	0.0979
VPact (mbar)	0.7795	1.1076	0.5095	0.3796	0.8860	1.3604	0.5180	0.4828	0.5169	0.4429	0.0444	0.0032
wd (deg)	62.790	7118.8	62.386	6700.7	62.150	6517.5	60.390	6418.9	60.635	6935.8	60.390	6480.8
wv (m/s)	1.8504	6.0069	1.0947	2.2187	2.2740	8.6357	1.6557	5.1354	1.7380	4.8712	1.0900	2.1452

Table 11

Prediction performance comparisons on *rain* (mm), *rh* (%), and *T* (degC) of Weather data. Best results are in **bold**. Second best results are underlined.

Weather measurement	rain (mm)		rh (%)		T (degC)	
	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓
Autoformer [26]	0.0233	0.0044	5.7257	55.650	1.5169	3.9537
Informer [25]	0.0207	0.0045	4.7953	38.448	1.4631	3.6560
TFT [46]	0.0100	0.0035	2.6100	<u>10.410</u>	1.3500	2.8400
FEDformer [27]	0.0230	0.0043	4.4780	34.108	0.9804	1.5919
N-HITS [42]	0.0163	0.0064	8.9792	170.35	1.5373	4.8461
TimesNet [37]	<u>0.0115</u>	<u>0.0042</u>	9.8202	173.08	2.2482	9.2797
<i>Ours - Autoformer-based</i>	0.0166	0.0043	3.6424	13.267	<u>0.5932</u>	<u>0.6377</u>
<i>Ours - Informer-based</i>	0.0177	0.0044	2.4865	10.540	1.1545	2.3377
<i>Ours - FEDformer-based</i>	0.0155	<u>0.0042</u>	0.6565	0.7572	0.1341	0.0343

Table 12

Prediction performance comparisons on *Tdew* (degC), *VPmax* (mbar), and *VPdef* (mbar) of Weather data. Best results are in **bold**. Second best results are underlined.

Weather measurement	Tdew (degC)		VPmax (mbar)		VPdef (mbar)	
	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓
Autoformer [26]	1.2092	2.5228	1.5275	3.8926	1.1536	2.5306
Informer [25]	1.1778	2.5829	2.1572	7.1303	1.2478	2.9590
TFT [46]	1.8800	5.2300	<u>0.3200</u>	<u>0.1500</u>	0.1000	0.0200
FEDformer [27]	0.7617	0.9600	1.0445	1.9000	0.8511	1.4185
N-HITS [42]	1.2951	3.1886	0.9997	2.1068	0.9216	1.9069
TimesNet [37]	1.5670	4.6231	1.9383	8.4014	1.8057	8.6581
<i>Ours - Autoformer-based</i>	0.8689	1.3016	1.1050	2.8171	1.0645	2.3658
<i>Ours - Informer-based</i>	<u>0.5399</u>	<u>0.4970</u>	1.9521	6.5402	0.5497	0.6393
<i>Ours - FEDformer-based</i>	0.0831	0.0142	0.1242	0.0282	<u>0.2206</u>	<u>0.0979</u>

Table 13

Performance of the proposed ConEm mechanism with three Transformer variants on *VPact* (mbar), *wd* (deg), and *wv* (m/s) of Weather data. Best results are in **bold**. Second best results are underlined.

Weather measurement	VPact (mbar)		wd (deg)		wv (m/s)	
	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓
Autoformer [26]	0.7795	1.1076	62.790	7118.8	1.8504	6.0069
Informer [25]	0.8860	1.3604	62.150	6517.5	2.2740	8.6357
TFT [46]	<u>0.1000</u>	<u>0.0200</u>	60.740	7189.8	1.1400	2.6900
FEDformer [27]	0.5169	0.4429	60.635	6935.8	1.7380	4.8712
N-HITS [42]	0.7429	1.1191	55.520	7775.5	1.3302	3.4352
TimesNet [37]	1.0046	2.0266	<u>59.298</u>	6748.7	1.1513	2.3205
<i>Ours - Autoformer-based</i>	0.5095	0.3796	62.386	6700.7	<u>1.0947</u>	2.2187
<i>Ours - Informer-based</i>	0.5180	0.4828	60.390	6418.9	1.6557	5.1354
<i>Ours - FEDformer-based</i>	0.0444	0.0032	60.390	<u>6480.8</u>	1.0900	2.1452

significant decreases in accuracy. The proposed Feature Tokenizer also plays a critical role, as its absence results in a performance decline, with MAE increasing by 9%→15% and MSE increasing by 6%→10% in short-term forecasting experiments.

4.4.2. Model complexity

We evaluate the computational overhead introduced by ConEm by comparing GPU memory consumption and training time for FEDformer and Informer across the benchmark EPF dataset (Table 18). For FEDformer, integrating ConEm increases peak GPU memory usage by only 1.7% (from 5245MB to 5332MB on average) and training time by 1.7% (from 5832s to 5931s), indicating minimal overhead. For Informer, GPU usage rises by 19.8% (from 626MB to 750MB) and training time by 5.2% (from 925s to 973s), which remains modest given the significant performance gains. Across all datasets, the memory and time increases are consistently small, with no evidence of instability or major bottlenecks. These results confirm that ConEm offers substantial forecasting improvements with negligible additional computational cost, making it suitable for practical deployment in real-world forecasting systems.

5. Discussions and future works

In this paper, we define external factors as all known variables within our scope, while excluding unexpected factors, such as environmental disasters, from our application. Exploring methods to incorporate unknown factors without historical data into time series forecasting presents an interesting and challenging future direction. It can be observed that external factors derived from meteorological experiments exert a more substantial influence on the enhancement of forecasting accuracy compared to external factors from sales forecasting experiment. One possible reason is that external factors from sales forecasting experiment are sparse in nature such as promotion campaign. Moreover, some external factors need more data collection to model its impact on the target feature. For example, promotion factor is only stored as binary feature if item is in promotion for Corporacion Favorita Grocery Sales dataset, while promotion information includes the price change, the marketing strategy, etc. Another possible reason is that weather conditions are usually highly-correlated in nature as seen in Fig. 6, and its data collection is sufficient with frequent update. Furthermore, experimenting with the development of foundation models based on external factors could be advantageous, especially as these models have recently seen widespread application in both academic and industrial settings.

As discussed in the complexity analysis, our mechanism's computational cost scales with the feature dimensionality. This introduces potential bottlenecks and memory constraints, particularly when handling a large number of external factors in high-dimensional dataset. The inclusion of multiple contextual embeddings increases the

Table 14

Forecasting error comparison on the EPF dataset for Autoformer [26], Informer [25], and FEDformer [27], with and without the integration of the ConEm mechanism.

Dataset	Autoformer [26]		+ ConEm		Informer [25]		+ ConEm		FEDformer [27]		+ ConEm	
	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓
NP	0.402	0.398	0.325	0.355	3.420	1.560	0.156	0.228	0.321	0.360	0.191	0.272
PJM	0.168	0.267	0.112	0.212	0.224	0.316	0.105	0.183	0.199	0.297	0.124	0.237
BE	0.500	0.333	0.439	0.315	0.668	0.481	0.387	0.251	0.493	0.335	0.436	0.309
FR	0.519	0.295	0.483	0.260	0.644	0.398	0.423	0.211	0.511	0.284	0.491	0.270
DE	0.674	0.544	0.465	0.452	2.339	1.235	0.314	0.353	0.807	0.587	0.464	0.461
AVG	0.453	0.367	0.364	0.319	1.459	0.798	0.277	0.245	0.467	0.373	0.341	0.309

Table 15

Forecasting error comparison on the EPF dataset for PatchTST [48] and iTransformer [49], with and without the ConEm mechanism. Two integration strategies are evaluated: (i) embedding ConEm directly within the encoder only (+ ConEm (E)), and (ii) appending ConEm to both the encoder and the encoder output before the linear forecasting layer (+ ConEm).

Dataset	PatchTST [48]		+ ConEm (E)		+ ConEm		iTransformer [49]		+ ConEm (E)		+ ConEm	
	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓
NP	0.2311	0.2785	0.2267	0.2734	0.1785	0.2348	0.2257	0.2741	0.2232	0.2705	0.2051	0.2615
PJM	0.1469	0.2498	0.1364	0.2355	0.0992	0.2033	0.1430	0.2333	0.1412	0.2306	0.1180	0.2118
BE	0.3860	0.2695	0.3735	0.2592	0.3723	0.2509	0.4126	0.2799	0.4073	0.2777	0.3971	0.2686
FR	0.4433	0.2419	0.4350	0.2384	0.4086	0.2138	0.4875	0.2472	0.4802	0.2449	0.4692	0.2398
DE	0.4926	0.4480	0.4560	0.4368	0.3410	0.3709	0.4619	0.4434	0.4528	0.4345	0.3905	0.3972
AVG	0.33998	0.29754	0.32552	0.2887	0.2799	0.2547	0.3461	0.29558	0.34094	0.29164	0.3160	0.2758

Table 16

Forecasting error comparison on the EPF dataset for various baseline models and models integrated with ConEm.

Model	NP		PJM		BE		FR		DE		AVG	
	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓
Autoformer [26]	0.402	0.398	0.168	0.267	0.500	0.333	0.519	0.295	0.674	0.544	0.453	0.367
Informer [25]	3.420	1.560	0.224	0.316	0.668	0.481	0.644	0.398	2.339	1.235	1.459	0.798
FEDformer [27]	0.321	0.360	0.200	0.297	0.493	0.335	0.511	0.284	0.807	0.587	0.467	0.373
SCINet [36]	0.373	0.368	0.143	0.259	0.731	0.412	0.855	0.384	0.565	0.497	0.533	0.384
DLinear [44]	0.309	0.321	0.108	0.215	0.463	0.313	0.429	0.260	0.520	0.463	0.366	0.314
TimeNet [37]	0.250	0.289	0.097	0.195	0.419	0.288	0.431	0.234	0.502	0.446	0.340	0.290
TiDE [62]	0.335	0.340	0.124	0.228	0.523	0.336	0.510	0.290	0.568	0.496	0.412	0.338
Crossformer [28]	0.240	0.285	0.101	0.199	0.420	0.290	0.434	0.208	0.574	0.430	0.354	0.284
PatchTST [48]	0.231	0.279	0.147	0.250	0.386	0.270	0.443	0.242	0.493	0.448	0.340	0.298
RLinear [50]	0.335	0.340	0.124	0.229	0.520	0.337	0.507	0.290	0.547	0.498	0.412	0.339
iTransformer [49]	0.265	0.300	0.097	0.197	0.394	0.270	0.439	0.233	0.479	0.443	0.335	0.289
TFT [46]	0.369	0.391	0.141	0.241	0.479	0.305	0.461	0.249	0.559	0.490	0.402	0.335
Stationary [61]	0.294	0.308	0.122	0.228	0.433	0.289	0.466	0.242	0.483	0.447	0.360	0.303
NBEATSx [60]	0.272	0.301	0.097	0.189	0.389	0.265	0.393	0.211	0.499	0.447	0.330	0.283
TimeXer [31]	0.236	0.268	0.093	0.192	0.379	0.243	0.385	0.208	0.440	0.415	0.307	0.265
<i>Ours - Autoformer-based</i>	0.325	0.355	0.112	0.212	0.439	0.315	0.483	0.260	0.465	0.452	0.364	0.319
<i>Ours - Informer-based</i>	0.156	0.228	0.105	0.183	0.387	0.251	0.423	0.211	0.314	0.353	0.277	0.245
<i>Ours - FEDformer-based</i>	0.191	0.272	0.124	0.237	0.436	0.309	0.491	0.266	0.464	0.461	0.341	0.309
<i>Ours - PatchTST-based</i>	0.179	0.235	0.099	0.203	0.372	0.251	0.409	0.214	0.341	0.371	0.280	0.255
<i>Ours - iTransformer-based</i>	0.205	0.262	0.118	0.212	0.397	0.269	0.469	0.240	0.391	0.397	0.316	0.276

Table 17Long-term forecasting results for T (degC) and rh (%) on Weather. Best results are in **bold**. Second best results are underlined.

Model	T (degC)		rh (%)	
	MAE	MSE	MAE	MSE
w/o LoEm	0.1520	0.0403	0.6639	0.7743
w/o FuEm	2.7054	12.534	11.746	230.01
w/o PaEm	1.0045	1.6266	3.7893	24.187
w/o Feature Tokenizer	1.8743	6.3456	4.2125	45.218
<i>Full</i>	0.1341	0.0343	0.6565	0.7572

number of input tokens the model must attend to, potentially raising memory consumption. However, by separating these contexts, each em-

bedding is able to specialize in capturing distinct aspects of the data-such as static features, the historical (look-back) window, and the prediction horizon-thereby simplifying the learning of a fused representation. To address scalability in extremely high-dimensional settings, future work may explore dimensionality reduction techniques, attention sparsity or parallel processing mechanisms to optimize memory and computational efficiency. In addition, while the primary focus of our framework is on improving predictive performance and preserving architectural flexibility, we acknowledge that comprehensive interpretability is beyond the current scope of this work. As part of future research, our framework could be extended with explicit interpretability modules-such as attention weight visualizations or feature importance analyses-to provide deeper insights across various datasets.

Table 18

Comparison of peak GPU memory usage (MB) and training time (seconds) for FEDformer [27] and Informer [25] with and without ConEm across EPF dataset.

Dataset	FEDformer [27]		+ ConEm		Informer [25]		+ ConEm	
	GPU (MB)	Training (s)	GPU (MB)	Training (s)	GPU (MB)	Training (s)	GPU (MB)	Training (s)
NP	5246	7432	5345	7563	623	902	735	936
PJM	5235	3875	5336	3932	612	926	712	981
BE	5265	4432	5331	4516	648	946	778	992
FR	5235	4861	5324	4952	612	937	742	982
DE	5246	8561	5323	8693	634	912	785	975
AVG	5245	5832	5332	5931	626	925	750	973

6. Conclusion

Our paper have presented a novel solution to address the existing limitations in incorporating external factors into short- and long-term time series forecasting. The proposed ConEm offers a novel paradigm for modeling and embedding the impact of exogenous features by integrating contextual embeddings with temporal dependencies, enhancing the forecasting performance for many encoder-decoder frameworks. Finally, comprehensive real-world experiments on Weather, Retail and Energy datasets showed that backbone models with ConEm outperformed both their counterparts without ConEm and other SOTA models.

CRedit authorship contribution statement

Hoang Nguyen Nguyen: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Wei Xi-ang:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization; **Lianhua Chi:** Writing – review & editing; **Mike Da Gama:** Writing – review & editing, Validation, Project administration, Data curation; **Sanjeevani Avashi:** Validation, Data curation; **Michael Treloar:** Validation, Data curation; **Lu Yu:** Writing – review & editing.

Data availability

I have shared the code and data links. Australian Pharmaceutical Weekly Dispensary data and Australian Daily POS data cannot be shared due to the confidential business and customer information they contain.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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